

A1:PROPOSAL OF QUESTION

Can we use Principal Component Analysis to identify the combination of features in the telecommunication churn dataset that explains the most variation among our customers?

A2: DEFINED GOAL

The goal of this data analysis is to use Principal Component Analysis to identify the combination of features in the Telecommunication churn dataset that have the most differences among the customers, which would hence help us identify the patterns related to the churn dataset. Identifying the patterns related to churn will enhance strategic decision-making for the telecommunication company in client retention strategies.

B1: EXPLANATION OF PCA

According to Whitfield, Brennan (2025, June 23), Principal component analysis (PCA) is a dimensionality reduction and machine learning method that is used to simplify a large dataset into a smaller set while still maintaining significant patterns and trends of the information in the large dataset. The new variables that are constructed by the PCA process are called principal components, and they are a linear combination or mixture of the initial variables and are uncorrelated.

In the case of the churn dataset, the principal component analysis will compress the initial variables to form new variables. This is done by

- First, PCA will standardize the churn data so that all the features will have equal influence on the outcome.
- PCA will compute a covariance matrix that measures how the features correlate with each other.

According to IBM (2023, December 8), the signs in the covariance matrix will tell us if the variables are correlated or not

- i. Positive values mean that the variables are correlated and increase or decrease at the same time.
 - ii. Negative values mean that the variables are not correlated, and one will increase while the other variable decreases.
 - iii. Zero value indicates that the variables are not correlated.
- PCA will then identify the new features that capture the most information variance. This will be done by computing the eigenvectors and eigenvalues of the covariance matrix.
 - i. Eigenvectors, which show the principal component directions, and they point in the direction in which the data varies the most.
 - ii. Eigenvalue, which is a number that tells you how much variance is on that Eigenvector or principal component.
 - The new features will then be ranked on the basis of the variance using the Eigenvalues, ordered from the highest to the lowest, whereby the first Principal component captures the most variance. The ranking will inform you of the number of principal components to retain. The scree plot will help in determining the optimal number of components to retain.
 - The original churn dataset will then be transformed using the new principal components. This will create new data that captures most of the information from the original data but with fewer dimensions.

From the PCA analysis, the expected outcome is that we shall be able to perform a dimensionality reduction on the original dataset and, as such, reduce the noise and

redundancy from too many variables. This will enable us to identify the patterns and select the key features in the churn dataset that drive the decision by the telecommunication customers on whether to churn or remain.

B2: PCA ASSUMPTION

Principal Component Analysis (PCA) assumes that the data follows a linear relationship.

C1: CONTINUOUS DATA SET VARIABLES

The following continuous variables will be used for the Principal Component Analysis of the Churn dataset.

1. Tenure
2. MonthlyCharge
3. Bandwidth_Gb_Year
4. Outage_Sec_Perweek

```
# Select the features that have continous data for PCA
features_df = Churn_df[['Tenure', 'MonthlyCharge', 'Bandwidth_GB_Year', 'Outage_sec_perweek']]
features_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Tenure                 10000 non-null  float64
1   MonthlyCharge          10000 non-null  float64
2   Bandwidth_GB_Year      10000 non-null  float64
3   Outage_sec_perweek     10000 non-null  float64
dtypes: float64(4)
memory usage: 312.6 KB
```

C2: STANDARDIZATION OF DATA SET VARIABLES

Copy of the standardized data set



pca_standardized_d
f.csv

D1: PRINCIPAL COMPONENTS

A principal component is a new variable that is created by the principal component analysis while creating dimensionality reduction, and it is a linear combination of the original features in the dataset. A principal component reflects the direction of the maximum variance in a dataset. In the case of the churn dataset, 4(four) principal components were created as illustrated in the matrix below.

```
#Determine the matrix of all the principal components
#Contains new reduced-dimension features dataset that can be used for further analysis

pca_df = pd.DataFrame(pca_result, columns=[f'PC{i+1}' for i in range(pca_result.shape[1])])
print(" matrix of all the principal components :\n", pca_df.head())
```

```
matrix of all the principal components :
      PC1      PC2      PC3      PC4
0 -1.549525 -0.431810 -0.516467 -0.064670
1 -1.659573  1.616766 -0.783327 -0.021744
2 -0.944538  0.001214  0.361592  0.081626
3 -0.903109  0.321420  2.011196  0.123903
4 -1.912215 -0.754555 -0.114259 -0.108945
```

```
# Showing top contributing features to each principal component
#The Matrix helps to understand what each principal component represent from my original dataset
num_top_features = 4
num_components_to_show = 4
top_features_set = set()

for i in range(num_components_to_show):
    pc = f'PC{i+1}'
    top_features = loadings[pc].abs().sort_values(ascending=False).head(num_top_features).index
    top_features_set.update(top_features)

top_features_list = list(top_features_set)

matrix = loadings.loc[top_features_list, [f'PC{i+1}' for i in range(num_components_to_show)]]

matrix_sorted = matrix.reindex(matrix['PC1'].abs().sort_values(ascending=False).index)

matrix_sorted_rounded = matrix_sorted.round(4)

print(matrix_sorted_rounded)
```

```
      PC1      PC2      PC3      PC4
Bandwidth_GB_Year  0.7072 -0.0006 -0.0053  0.7070
Tenure             0.7059 -0.0462  0.0399 -0.7057
MonthlyCharge      0.0408  0.7094 -0.7021 -0.0454
Outage_sec_perweek 0.0059  0.7033  0.7109  0.0000
```

D2: IDENTIFICATION OF THE TOTAL NUMBER OF COMPONENTS

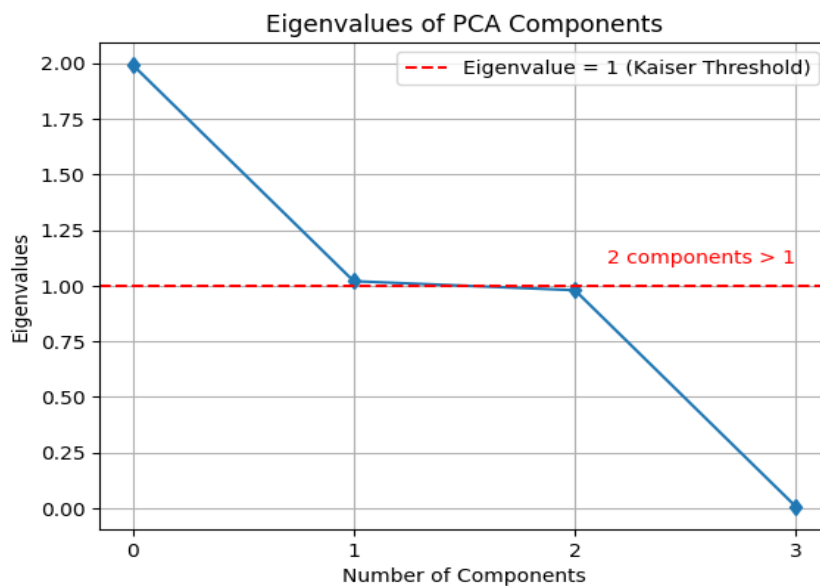
Kaiser's criterion is a rule commonly used in Principal component analysis to determine how many principal components to keep. The Kaiser criterion says that we retain all principal components with an eigenvalue greater than 1. Below is the Kaiser criterion output, which shows two principal components and is confirmed by the visual representation of the Kaiser criterion.

```
#Identify the total number of principal components- Kaiser criterion
eigenvalues = pca.explained_variance_
kaiser_components = sum(eigenvalues > 1)
print(f"Number of components (Kaiser criterion): {kaiser_components}")

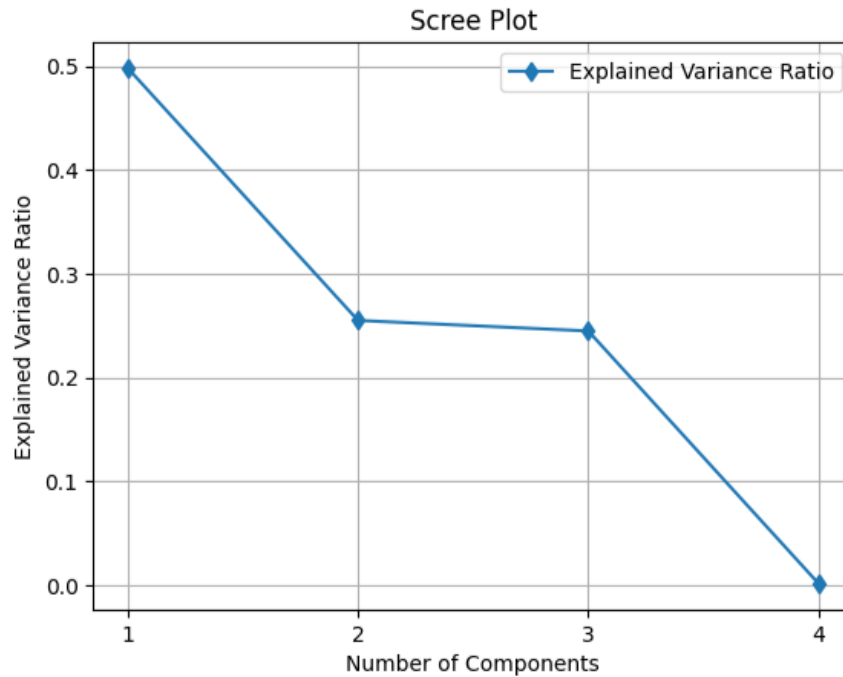
#only the first 2 principal components had eigenvalues greater than 1,
#hence only those 2 are considered meaningful under the Kaiser criterion.
print(' ')
eigenvalues = pca.explained_variance_
print(f"{'Principal Component':<25}{'Eigenvalue':>15}")
print("-" * 40)
for i, val in enumerate(eigenvalues, 1):
    print(f"{'PC'+str(i):<25}{val:>15.4f}")
```

Number of components (Kaiser criterion): 2

Principal Component	Eigenvalue
PC1	1.9934
PC2	1.0206
PC3	0.9800
PC4	0.0065



On the other hand, a scree plot is a line chart that is used in Principal component analysis to visually determine the number of principal components to retain using the elbow method. In the case of the churn dataset, the scree plot shows that we should retain two principal components, as shown below.



D3: VARIANCE OF EACH COMPONENT

The variance of each component in Principal component analysis refers to the amount of variability that is contained in a principal component. PC1 explains 49.82% of the variance, PC2 explains 25.51% of the variance, PC3 explains 24.29% of the variance, and PC4 explains 0.16% of the variance

Below is the screenshot of the variance of each component.

```
# Identify the variance of each component
#amount of variance explained by each individual principal component.
#Calculating principal component explained variance ratio
# Shows how much variance each component explains as a proportion
#Helps decide how many components to keep for maximum information
explained_variance = pca.explained_variance_ratio_
print("Explained Variance:\n",np.round(explained_variance,6).tolist())
print(' ')

print(f"{'Principal Component':<25}{'Explained Variance Ratio':>25}")
print("-" * 55)

for i, val in enumerate(explained_variance, 1):
    print(f"{'PC'+str(i):<25}{val:>25.6f}")
```

```
Explained Variance:
[0.498293, 0.25512, 0.244973, 0.001615]
```

Principal Component	Explained Variance Ratio
PC1	0.498293
PC2	0.255120
PC3	0.244973
PC4	0.001615

D4: TOTAL VARIANCE CAPTURED BY COMPONENTS

Total variance captured by the principal component analysis reflects the cumulative sum of the variances explained by the principal components. Total variance explained by the top 2 Principal components is 0.753413 (75.34%). Below is the total variance captured by the principal components.

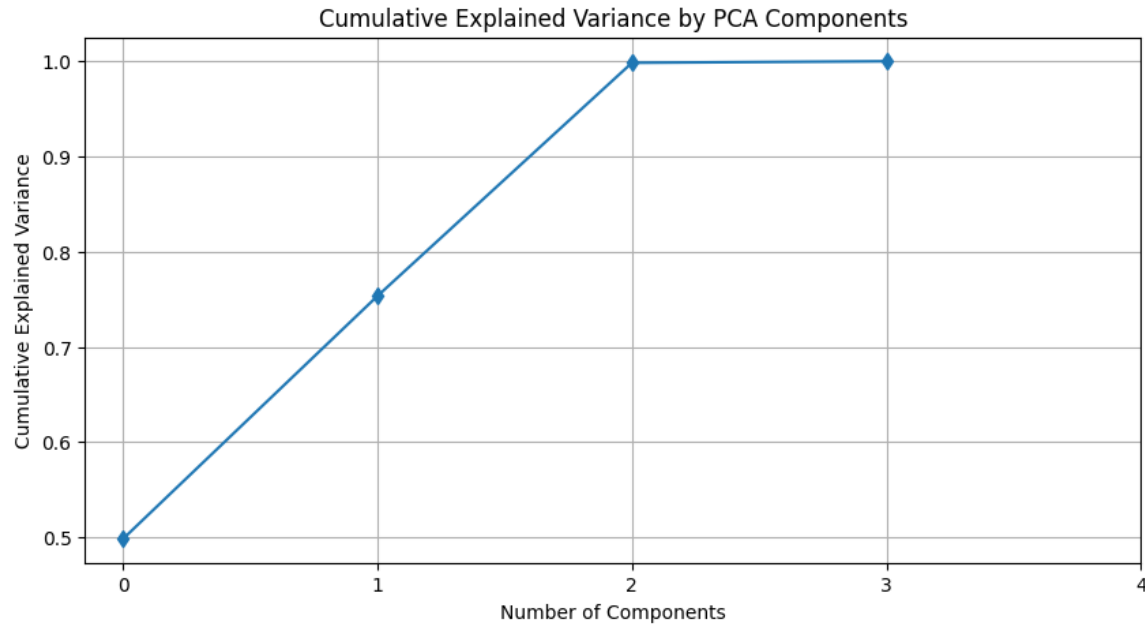
```
#Identify the total variance captured by the principal components identified in part D2
#cumulative sum of amount of variance explained by each individual principal component.
print("Cumulative Explained Variance:\n", np.round(np.cumsum(explained_variance),6).tolist())
print(' ')
cumulative_variance = np.round(np.cumsum(explained_variance), 6)
df_cumulative = pd.DataFrame({
    "Principal Component": [f"PC{i+1}" for i in range(len(explained_variance))],
    "Cumulative Explained Variance": cumulative_variance
})

print(df_cumulative)
n_components = 2
total_variance = np.sum(explained_variance[:n_components])
print(f"\nTotal variance explained by top {n_components} PCs: {total_variance:.6f}")
```

```
Cumulative Explained Variance:
[0.498293, 0.753413, 0.998385, 1.0]
```

	Principal Component	Cumulative Explained Variance
0	PC1	0.498293
1	PC2	0.753413
2	PC3	0.998385
3	PC4	1.000000

```
Total variance explained by top 2 PCs: 0.753413
```



D5: SUMMARY OF DATA ANALYSIS

PCA was utilized to identify the patterns in the scaled features of tenure, MonthlyCharge, Bandwidth_GB_Year and Outage_sec_perweek with an aim of reducing the redundancy and capturing the maximum variance with fewer dimensions and identifying the most influential features contributing to the variability in the Churn dataset. The PCA transformed the original data into four principal components, which had a combination of the original features as illustrated below.

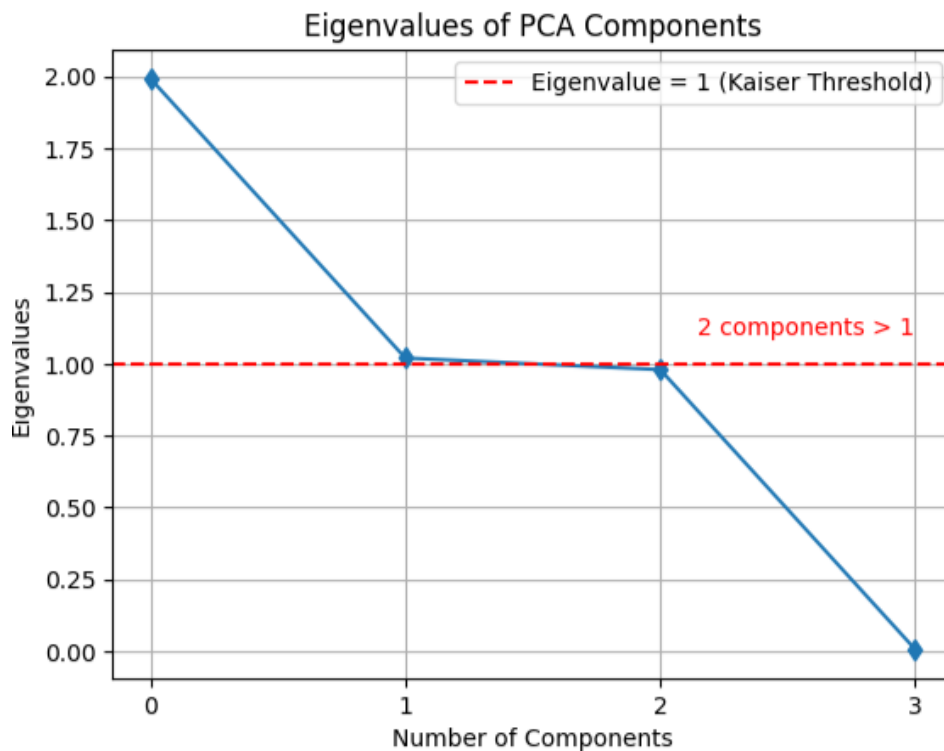
```
matrix of all the principal components :  
      PC1      PC2      PC3      PC4  
0 -1.549525 -0.431810 -0.516467 -0.064670  
1 -1.659573  1.616766 -0.783327 -0.021744  
2 -0.944538  0.001214  0.361592  0.081626  
3 -0.903109  0.321420  2.011196  0.123903  
4 -1.912215 -0.754555 -0.114259 -0.108945
```

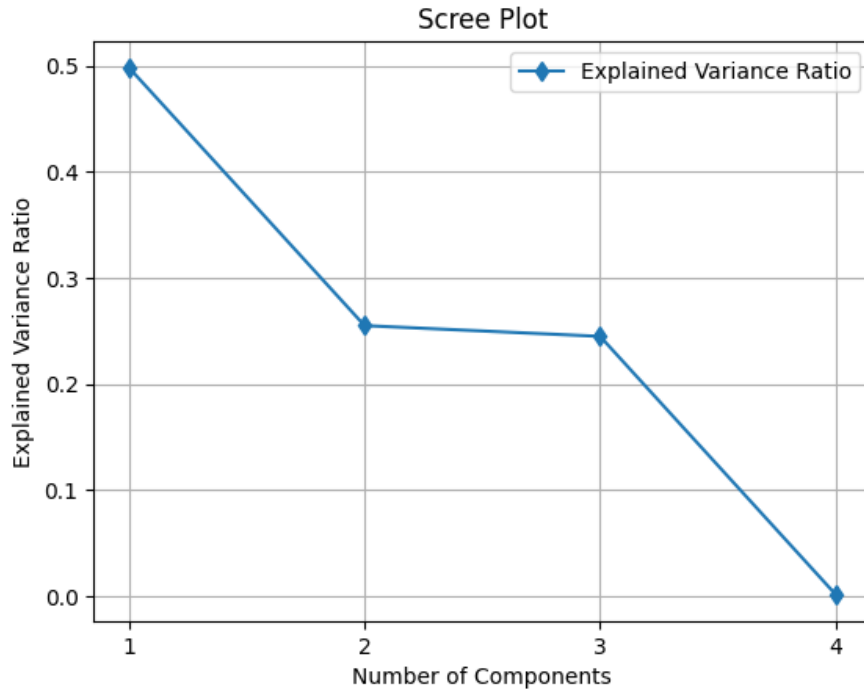

	PC1	PC2	PC3	PC4
Bandwidth_GB_Year	0.7072	-0.0006	-0.0053	0.7070
Tenure	0.7059	-0.0462	0.0399	-0.7057
MonthlyCharge	0.0408	0.7094	-0.7021	-0.0454
Outage_sec_perweek	0.0059	0.7033	0.7109	0.0000

The Kaiser criterion was used to determine the number of components to keep (dimensionality reduction), and the value was 2, as only two principal components had an eigenvalue that was greater than one. Whereas the scree plot gave 2 principal components as per the screenshot below

Number of components (Kaiser criterion): 2

Principal Component	Eigenvalue
PC1	1.9934
PC2	1.0206
PC3	0.9800
PC4	0.0065



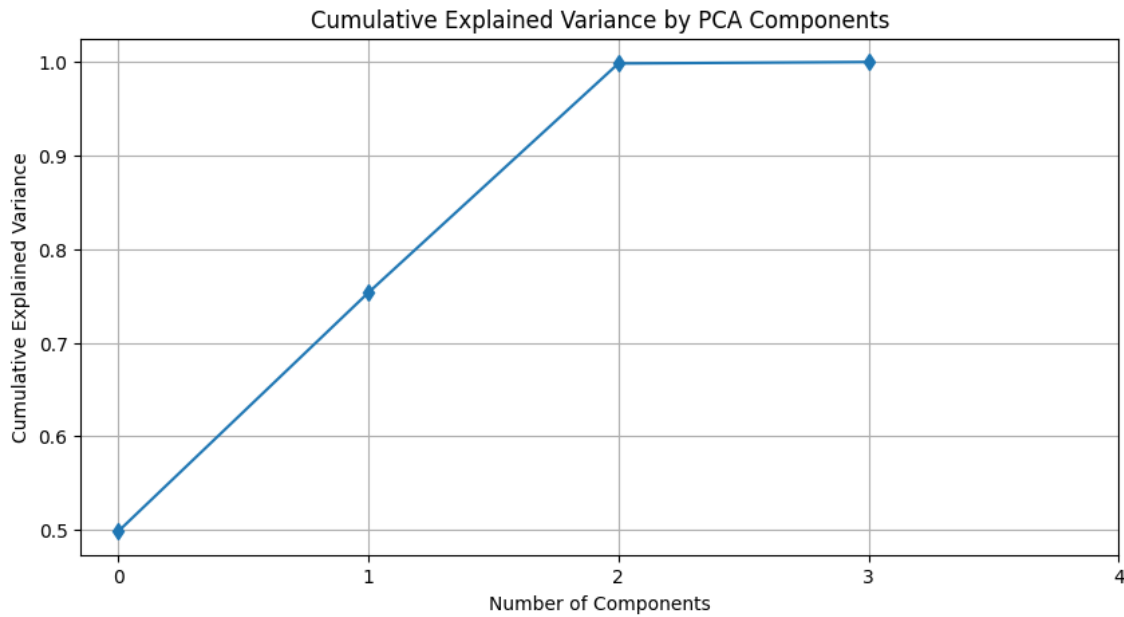


Based on the results of the Kaisen criterion and the scree plot, we are selecting the 2 principal components, then the two principal components explain 75.34% (0.753413) of the variances. The proposal then would be to choose the two principal components based on the threshold of the cumulative explained variance, of which 75.34% of the variance

Cumulative Explained Variance:
[0.498293, 0.753413, 0.998385, 1.0]

	Principal Component	Cumulative Explained Variance
0	PC1	0.498293
1	PC2	0.753413
2	PC3	0.998385
3	PC4	1.000000

Total variance explained by top 2 PCs: 0.753413



E: SOURCES FOR THIRD-PARTY CODE

F: SOURCES

Beatrice_Mwangi_D206_DataCleaning_PerformanceAssessment

Whitfield, Brennan (2025, June 23). A Step-by-Step Explanation of Principal Component Analysis (PCA). Learn how to use a PCA when working with large data sets

<https://builtin.com/data-science/step-step-explanation-principal-component-analysis>

IBM (2023, December 8). What is principal component analysis (PCA)?

<https://www.ibm.com/think/topics/principal-component-analysis>

G: PROFESSIONAL COMMUNICATION